📊 Adversarial ML — Results Summary

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Model: SOCAlertNet (Feed-Forward Neural Network)
Dataset: Synthetic SOC Alert Dataset (10,000 samples)

Attack vs Defence Comparison Table

Attack	Clean Model	Adversarially Trained Model
No attack (baseline)	**94.2%**	**91.8%**
FGSM ε=0.05	58.3%	86.1%
FGSM ε=0.10	31.7%	78.4%
FGSM ε=0.20	18.2%	64.9%
PGD ε=0.05 (40 steps)	29.4%	82.3%
PGD ε=0.10 (40 steps)	11.3%	71.2%
PGD ε=0.20 (40 steps)	5.1%	53.7%

Key Findings

1. Standard Models Are Extremely Vulnerable

A SOC alert classifier with **94.2% clean accuracy** can be reduced to just **11.3% accuracy** under PGD attack with ε=0.10. This means an attacker who knows the model architecture can craft alerts that almost always bypass detection.

2. FGSM is Fast but PGD is Devastating

- FGSM (single step) reduces accuracy to **31.7%** at ε=0.10
- PGD (40 steps) reduces accuracy to **11.3%** at ε=0.10 — 3x more effective
- This shows why PGD is considered the "gold standard" adversarial attack

3. Adversarial Training Works

After adversarial training:

- Clean accuracy drops slightly: 94.2% → 91.8% **(only -2.4%)**
- PGD ε=0.10 accuracy recovers: 11.3% → 71.2% **(+59.9%)**
- The small clean accuracy tradeoff is worth the massive robustness gain

4. Perfect Robustness Doesn't Exist

Even with adversarial training, accuracy under strong PGD (ε=0.20) drops to **53.7%** — better than 5.1% but still degraded. This is a fundamental limitation of current defences.

Security Implications for SOC/ML Teams

Risk	Impact	Mitigation
Adversarial alert crafting	Bypass ML-based detection	Adversarial training
Model extraction	Attacker learns your model	API rate limiting
Data poisoning	Degrade model over time	Data validation, monitoring
Gradient masking bypass	False sense of security	Certified defences

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Recommended Defences (Beyond Adversarial Training)

1. **Input Validation** — Reject inputs outside expected ranges
2. **Model Ensembles** — Harder to attack multiple models simultaneously
3. **Certified Defences** — Randomised smoothing provides provable robustness
4. **Anomaly Detection on Inputs** — Flag unusual alert feature distributions
5. **Human-in-the-loop** — Never fully automate high-stakes security decisions

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How to Reproduce

```bash
# 1. Train baseline model
python src/train_clean.py

# 2. Run FGSM attack
python src/attacks/fgsm.py --epsilon 0.1

# 3. Run PGD attack (stronger)
python src/attacks/pgd.py --epsilon 0.1 --steps 40

# 4. Apply adversarial training defence
python src/defence/adversarial_training.py

# 5. Full comparison
python src/evaluate.py
```

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